RAKESH CHOWDARY POLINA

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PROFILE

Experienced Software Engineer with a proven track record in developing scalable microservices, implementing robust facial recognition pipelines, and delivering cloud-native solutions. Skilled in Python, Java, Spring Boot, Flask, React.js, and AWS, with expertise in containerization, CI/CD, and Agile methodologies. Adept at optimizing performance, enhancing user engagement, and driving operational efficiency in production environments.

EDUCATION

Master of Science in Computer Science **May 2024**
Oklahoma State University, Stillwater, OK **CGPA: 3.909/4**

SKILLS

Languages:	Java, JavaScript, Python, SQL, C, C++, HTML, CSS, Swift
Frameworks:	React JS, Angular JS, Spring boot, Node JS, Hibernate, Spring MVC, React Native
Databases:	MySQL, MongoDB, NoSQL, PostgreSQL, Cassandra
Web Technologies:	RESTful API, Spring JDBC, JPA, OAuth 2, jQuery, JSP, JMS, RabbitMQ
Web Services:	REST, SOAP
Tools/Platforms:	Docker, Git, JIRA, Jenkins, Postman, Ansible, Kubernetes, Apache Kafka, Terraform, Swagger
AWS Cloud Services:	EC2, S3, EKS, RDS, ELB, Cloud Watch, CloudFront
Build Tools:	Maven, Gradle

WORK EXPERIENCE

Research Data Analyst, Oklahoma State University **May 2023 – May 2024**

- Engineered and implemented a robust facial recognition pipeline, performing both quantitative and qualitative analysis on image data with Python and InsightFace, to achieve a precision rate of 93.6%
- Implemented DBSCAN clustering and rigorously verified data accuracy across 182,000+ images, enhancing archive reliability and insight precision.
- Designed an intuitive UI for the labeling tool using Flask and React.js, streamlining user engagement and annotations.
- Designed and maintained database structures using SQL, enhancing user interaction management and feedback data integrity, ensuring operational excellence.

Software Engineer, Ericsson, India **July 2019 – July 2022**

- Modernized legacy monolithic applications to cloud-native microservices using Spring Boot and RESTful APIs, improving scalability to handle 50% more concurrent users.
- Migrated on-prem solutions to AWS (EC2, RDS, S3) by refactoring Java services, integrating Elastic Load Balancer and Auto Scaling for high availability.
- Implemented event-driven architecture with Apache Kafka for asynchronous communication, reducing system coupling and enhancing fault tolerance.
- Developed full-stack solutions using Java/Spring Boot for back-end services and React for dynamic front-end interfaces. Designed REST/GraphQL APIs with Spring WebFlux for reactive, non-blocking performance, reducing latency by 40%.
- Utilized Spring Data JPA and PostgreSQL for efficient data modeling, optimizing complex SQL queries to cut query execution time by 25%.
- Built CI/CD pipelines with Jenkins and Git, automating deployments to AWS EKS clusters. Integrated SonarQube for security scanning and code quality checks, reducing vulnerabilities by 30%.
- Containerized applications using Docker and orchestrated with Kubernetes, cutting deployment time by 50% and enabling zero-downtime releases.
- Implemented event-based integrations using Kafka for real-time data streaming between microservices. Leveraged Spring Security with OAuth2 for role-based access control, reducing unauthorized access incidents by 35%.
- Conducted end-to-end testing with JUnit/Mockito, achieving 95% test coverage, and used Postman for API contract validation.
- Collaborated with Agile teams using Jira/Confluence for sprint planning, backlog refinement, and documentation.
- Partnered with DevOps to monitor production systems via New Relic and Splunk, resolving critical issues 20% faster. Delivered bi-weekly releases aligned with sprint goals, improving feature delivery consistency by 25%.
- Designed PostgreSQL schemas for high-volume transactional systems, optimizing indexes and partitioning strategies. Integrated React front-end components with Spring MVC for server-side rendering, enhancing UI load times by 30%.
- Troubleshot live production issues using SLF4J logging and AWS CloudWatch, reducing downtime by 30%. Refactored legacy Java EE codebases to Spring Boot, eliminating deprecated dependencies and improving maintainability.